

Supplementary information

Scoring rendered charts by recovery of two-sample differences

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S1. Scope and status of this revision

This Supplement describes the pixel-recovery development benchmark. Earlier compression results are archived in the reproducibility package and are not evidence for the current score. The current method uses 21 settings from seven implemented chart families after screening a 15-family catalogue. It has no human-reader data, held-out observer validation or universal chart-ranking claim. All source arrays remain unchanged. The study protocol was fixed for the reported computational run after implementation checks, not preregistered before development.

S2. Complete catalogue of 15 families

Catalogue identifiers remain fixed and are not quality ranks. The catalogue is a familiar set specified for this project; no worldwide usage-frequency claim is made. Eligibility here means that an explicit decoder for the same five marginal quantile differences is implemented. An absent contract is not a score of zero. Some other families can be repurposed, but doing so requires a declared transformation and a newly evaluated contract.

ID	Family	Usual task	Current implementation
1	Bar chart	Category values	Group means; one setting
2	Line chart	Ordered trends	No trend target or implemented marginal decoder
3	Scatter plot	Joint relationships	No joint-association target or decoder
4	Pie chart	Parts of a whole	No declared composition transformation or decoder
5	Histogram	Numerical distribution	Five common-bin settings
6	Box plot	Median, spread, extrema	Five-number summary; one setting
7	Heatmap	Matrix patterns	Two group-probability strips; five bin settings
8	Stacked bar	Totals and composition	No declared composition transformation or decoder
9	Area chart	Ordered magnitude	No ordered-trend target or decoder
10	Bubble chart	Relationships and magnitude	No joint target or bubble-area decoder
11	Dot plot	Observations or category values	Means and raw observations; two settings
12	Treemap	Hierarchical composition	No hierarchy, composition target or decoder
13	Choropleth	Geographic values	No geographic identifiers, target or decoder
14	Violin plot	Distribution shape	Six sampled-density settings
15	Interval plot	Variability or uncertainty	Five-number variability display; one setting; not an uncertainty interval

S3. Exact computational contract

Target. NumPy's default linear empirical quantiles at probabilities (.1, .25, .5, .75, .9) define five equally weighted differences, group B minus group A. The target is descriptive and marginal even when the source observations are paired. It is not a full-distribution metric or an estimate of individual change.

Normalization. Pooled within-group standard deviation weights each sample variance by its degrees of freedom. If that standard deviation is zero, the implementation uses one source unit and records a fallback flag. This special case is not unit-invariant. All eligible candidates within a comparison share the same scale.

Range. Let $h_g = \text{SD}(g)n_g^{-1/5}$ and $b = \max(h_x, h_y, .01s)$. The measurement range is $[\min(0, \min(x, y) - 4b), \max(0, \max(x, y) + 4b)]$. If needed, a degenerate upper limit is replaced by the lower limit plus one. All families use this range, including histogram and heatmap bins. Zero inclusion supports bars but may reduce useful resolution for other displays; it is an explicit design limitation.

Inputs to the decoder. RGB image, family, setting, panel resolution, image width, measurement limits, panel bounding coordinates and declared group colours. The decoder is not given samples, target quantiles, pre-rendered summary values, placement offset or jitter seed. It is given exact axes rather than recovering tick labels through optical character recognition. The chart type is known.

Raster geometry. Two panels of side $H \in \{128, 256, 512\}$ are separated by 16 pixels. Each plotting region extends from local horizontal coordinate 8 to $H - 9$ and vertical coordinate 8 to $H - 9$. RGB colours are (38, 126, 165) and (223, 120, 54). Images are first drawn at fourfold resolution, reduced with Lanczos filtering and blurred by a Gaussian radius of 0.5 target pixels. Measurement marks use vertical offsets 0, .25, .5 and .75 pixels. These placements are sensitivity conditions, not independent samples of reader performance.

Bars and dots. Mean bars have width $0.4H$ and extend from zero to the group mean. Mean dots have radius two target pixels. Raw dots have radius 1.5 pixels and uniformly jittered horizontal centres within $\pm 0.35H$ of the panel centre. Random seeds are $20260920 + r + 100003g$, with zero-based placement index r and group index g . Thus raw-dot realizations vary both placement and jitter. The dots are opaque; within-group overlaps are retained.

Boxes and intervals. A one-pixel minimum–maximum line is common to both. A box has width $0.4H$ and spans the interquartile range, with a one-pixel black median line. The interval has a five-pixel interquartile line and black median marker of radius 2.5 pixels. These are five-number summaries of observations, not standard errors, confidence intervals or Tukey outlier conventions.

Histograms and heatmaps. Bin counts 4, 8, 16, 32, 64 share equal-width measurement bins. Histogram heights represent group probabilities on a fixed zero-to-one axis. Heatmap tiles encode probability m as $(\text{round}(255(1 - m)), \text{round}(255(1 - m)), 255)$. The heatmap strip occupies the middle 30% of plot height; its raster is invariant across the four vertical placements. This known linear colour scale is an oracle supplied through the implemented decoder, not evidence of perceptual linearity.

Violins. Group-specific Gaussian kernel densities use SciPy’s Scott bandwidth and are sampled on common grids of 8, 16, 32, 64, 128, 256 points. Width is normalized separately by each group’s peak to $0.4H$. The rendered polygon connects those grid points; no finer smoothing is inserted into the displayed figures. Constant samples use a Gaussian width of .01s, evaluated with a stabilized exponential to avoid underflow.

Pixel extraction. A pixel is group-coloured if its Euclidean RGB distance from the known group colour is below 80. A black mark has RGB norm below .78 times the group-colour norm. Bars use rows with coloured width exceeding 20% of plot width, taking the endpoint farther from zero. Mean dots use the coloured row-mass centroid. Box and interval extrema use any coloured or black mark; interquartile bounds use row width above 30% of plot width for boxes or 2.5 pixels for intervals. Median position is the black row-mass centroid. Recovered five-number landmarks are monotonized and linearly interpolated to the target probabilities.

Histogram probabilities come from the top coloured pixel at each bin-centre column relative to the known baseline. Heatmap probabilities come from inverse red intensity at tile centres. Negative recovered masses are clamped to zero and masses normalized. Their cumulative inverse uses linear interpolation within each nonempty bin. Violin and raw-dot decoders instead invert cumulative coloured pixel area by measurement row. This is an area-based observer, not a point-counting algorithm or an optimal image decoder. Empty recoveries fail explicitly.

Score and optimization. For each realization, divide mean absolute contrast error by the shared scale. Average that loss over all four realizations, then transform by $100/(1 + L)$. Do not average scores before optimization. A setting with any failed realization is excluded. Within each family select its lowest mean loss; ties can be displayed deterministically but remain equal scientific outcomes. Across families, absolute loss differences at most 10^{-12} are treated as ties, with no relative tolerance. Unrounded values determine rankings. Displaying three decimals does not assert precision or practical significance.

S4. Mathematical interpretation and limitations

For finite nonnegative loss, $0 < S \leq 100$ and $dS/dL = -100/(1+L)^2 < 0$. Thus maximizing score exactly minimizes the stated recovery loss. A monotone logarithmic or reciprocal transformation cannot distinguish equal losses. Injecting chart-specific offsets would change the objective and cannot establish visual superiority.

Let $e_{g,k} = \tilde{Q}_g(p_k) - Q_g(p_k)$. Then the contrast error is $e_{y,k} - e_{x,k}$. Shared errors can cancel, yielding a high contrast score despite inaccurate group quantiles. Conversely, a sample’s biological separation may be small yet faithfully recovered. Score is therefore neither a measure of separation nor a diagnostic of data quality.

For population target Δ_k^{pop} , the triangle inequality gives

$$\frac{1}{5} \sum_k |\tilde{\Delta}_k - \Delta_k^{\text{pop}}| \leq \frac{1}{5} \sum_k |\tilde{\Delta}_k - \Delta_k| + \frac{1}{5} \sum_k |\Delta_k - \Delta_k^{\text{pop}}|.$$

Improving image recovery addresses the first term only. The current experiment does not estimate the second term or supply confidence coverage. Placement ranges are neither sampling intervals nor probabilities that a selected chart is correct.

With nonconstant samples and ideal arithmetic, multiplying all observations by a positive constant scales the range, recovered values and normalization together. The checks verify this for the stated test inputs. Translation invariance is not asserted because the axis includes zero. Invariance checks on finite fixtures do not constitute a proof of all floating-point behaviour.

S5. Data provenance and primary analysis

The nine primary resources contain 127 prepared comparisons: WDBC 30; Cleveland five; heart failure six; CKD 11; ILPD nine; retinopathy 16; HAR 40; BIDMC five; wrist five.^{1–9} HAR uses participant/activity medians for 30 people; BIDMC uses 46 patient-level summaries; wrist uses early–late reference summaries for eight participants.^{10;11} Retinopathy lacks verified participant linkage. The 22 Parkinsons features are retained only as exploratory data because parsed identifiers indicate 32 groups while source documentation states 31 participants.¹²

Two CKD potassium values, 39 and 47 at one-based source rows 62 and 129, remain unchanged; their correctness and units could not be resolved. Prior compression omission analyses are archived but are not claimed as sensitivity analyses of the present pixel score. Full acquisition records, checksums, exclusions and transformations appear in `DATA_PROVENANCE.md` and the prepared-data pipeline.

S6. Verification and reproducibility files

The executable checks cover all 21 settings using known numerical inputs. They test finite monotone quantile output, positive-unit rescaling, blank-image rejection, PNG/metadata round-trip recovery and response to changing one group’s pixels. They also restrict the decoder’s metadata keys to exclude samples and targets. These tests establish plumbing and numerical consistency, not human validity or optimality of the decoder.

File in <code>rendered_study/results/</code>	Contents
<code>render_trials.csv</code>	All 37,548 realized scores, losses and recovery failures
<code>configuration_scores.csv</code>	All 9,387 setting–resolution results, averaged over four placements
<code>family_scores.csv</code>	All 3,129 best-family results across comparisons and resolutions
<code>winners.csv</code>	All 447 winning sets and tie counts
<code>summary.json</code>	Primary and exploratory counts, failures and sensitivity summaries
<code>checks.json</code>	Executed software-check outcomes

Run `python reproduce_revision.py` from the package root to execute checks, benchmark and report. Dependencies are pinned in `requirements.txt`. The benchmark uses four worker processes and records deterministic seeds. To regenerate figures without rerunning the benchmark, execute `python rendered_study/report.py`. The Overleaf ZIP is a separate, minimal typesetting package; set `main.tex` as its main document and compile `supp.tex` separately for this Supplement.

Recovery failures in primary trials

Counts below refer to individual rendering realizations across all settings in each family; families have different numbers of settings. A setting must recover successfully in all four realizations to enter selection. A failure reflects this decoder’s inability to recover its required marks or positive mass under the stated rendering conditions. It does not establish that a human or a different decoder could not read the chart.

Family	128 pixels	256 pixels	512 pixels
Bar chart	26	20	16
Box plot	22	8	4
Dot plot	0	0	0
Heatmap	0	0	0
Histogram	0	0	0
Interval plot	0	0	0
Violin plot	20	20	20

S7. Prospective validation required

A reader study should specify the intended population and expertise strata, accessible displays, the five quantile-difference questions, randomization, exclusions and time limits before recruitment. Candidate settings and scores should be frozen using development data; evaluation should use untouched datasets. Primary human outcomes should be estimation error in source units and normalized units, with response time and confidence secondary. Repeated responses require participant and dataset dependence to be handled in the analysis. Sample size should follow a prospective power analysis for a prespecified practically meaningful difference. No participant count, approval, power result or measured reader benefit is asserted here. The included reader-study document remains a planning aid and must be reconciled with this frozen pixel protocol before use.

S8. Inspectable score arithmetic for illustrated optima

The five examples in main Figures 2 and 3 are reported here. The target differences are computed from observations; recovered differences come only from the raster decoder. Means below average the four realizations. The final column averages absolute error before normalizing, so it need not equal the absolute difference between the preceding two columns. Averaging the final column gives the loss used in the score. The CSV preserves every realization separately.

Breast tumour / Fractal dimension (worst): Violin plot; score 97.566.

Quantile	Target difference	Mean recovered	Mean normalized error
0.10	0.003109	0.002267	0.04923
0.25	0.0062125	0.0062177	0.00032
0.50	0.01048	0.011198	0.04197
0.75	0.017215	0.017249	0.00210
0.90	0.023668	0.023136	0.03113

Smartphone activity / Gyroscope-jerk variability: Histogram; score 94.960.

Quantile	Target difference	Mean recovered	Mean normalized error
0.10	0.38832	0.3821	0.06612
0.25	0.45916	0.4627	0.03760
0.50	0.51263	0.51889	0.06658
0.75	0.58777	0.59599	0.08745
0.90	0.74432	0.7436	0.00759

Heart disease / ST depression (oldpeak): Violin plot; score 92.837.

Quantile	Target difference	Mean recovered	Mean normalized error
0.10	0	0.13135	0.12474
0.25	0.55	0.50405	0.04363
0.50	1.2	1.0813	0.11268
0.75	1.475	1.4359	0.03710
0.90	1.8	1.7288	0.06764

Wrist exercise / Mean heart rate: Bar chart; score 95.348.

Quantile	Target difference	Mean recovered	Mean normalized error
0.10	5.5476	3.6526	0.06377
0.25	5.4159	3.6526	0.05934
0.50	5.1094	3.6526	0.04902
0.75	3.8326	3.6526	0.01721
0.90	2.0296	3.6526	0.05462

Bedside monitoring / Mean heart rate: Violin plot; score 98.019.

Quantile	Target difference	Mean recovered	Mean normalized error
0.10	-2.1296	-1.6093	0.04209
0.25	-0.98337	-1.1793	0.01585
0.50	-0.8654	-0.61579	0.02019
0.75	-0.27391	-0.44462	0.01381
0.90	-0.089397	-0.20159	0.00908

S9. Complete primary reference-resolution audit

All 127 primary comparisons are listed below. The full seven-family rankings and all 21 settings for each comparison are in the accompanying CSV files. Scores are computational and conditional; this table is not a clinical comparison of datasets.

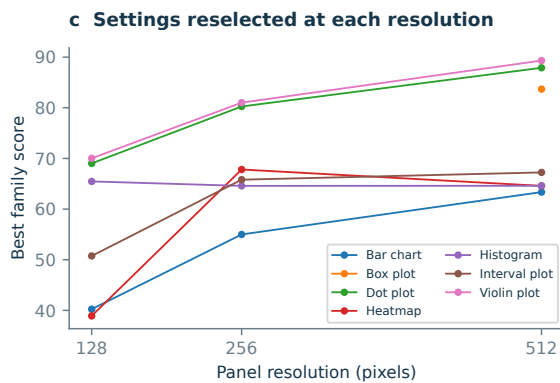
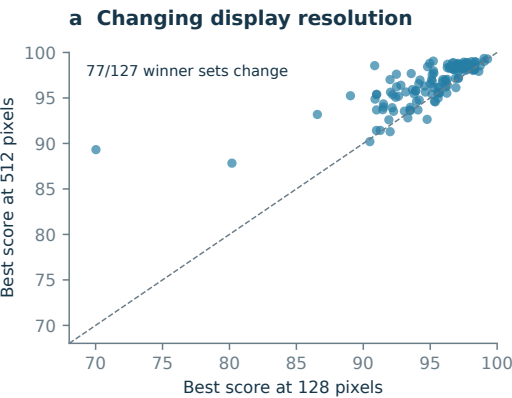
Key	Feature	Winning family set	Score
BIDMC__00	mean	Violin plot	98.019
BIDMC__01	median	Dot plot	98.361
BIDMC__02	q10	Histogram	98.062
BIDMC__03	q90	Dot plot	97.658
BIDMC__04	SD	Dot plot	98.253
CKD__00	age	Heatmap	98.680
CKD__01	bp	Dot plot	95.680
CKD__02	bgr	Heatmap	98.701
CKD__03	bu	Dot plot	98.325
CKD__04	sc	Heatmap	95.222
CKD__05	sod	Dot plot	94.243
CKD__06	pot	Dot plot	97.124
CKD__07	hemo	Dot plot	96.414
CKD__08	pcv	Heatmap	98.986
CKD__09	wc	Dot plot	96.156
CKD__10	rc	Heatmap	95.557
Cleveland__00	age	Heatmap	96.798
Cleveland__01	trestbps	Dot plot	93.880
Cleveland__02	chol	Heatmap	98.298
Cleveland__03	thalach	Violin plot	98.009
Cleveland__04	oldpeak	Violin plot	92.837
HAR__00	tBodyAcc-mean()-X	Heatmap	94.479
HAR__01	tBodyAcc-mean()-Y	Heatmap	96.176
HAR__02	tBodyAcc-mean()-Z	Dot plot	92.490
HAR__03	tBodyAcc-std()-X	Histogram	91.676

Key	Feature	Winning family set	Score
HAR__04	tBodyAcc-std()-Y	Heatmap	95.525
HAR__05	tBodyAcc-std()-Z	Dot plot	92.906
HAR__06	tGravityAcc-mean()-X	Histogram	93.457
HAR__07	tGravityAcc-mean()-Y	Heatmap	96.559
HAR__08	tGravityAcc-mean()-Z	Histogram	94.442
HAR__09	tGravityAcc-std()-X	Violin plot	81.015
HAR__10	tGravityAcc-std()-Y	Violin plot	86.545
HAR__11	tGravityAcc-std()-Z	Violin plot	87.882
HAR__12	tBodyAcc.Jerk-mean()-X	Dot plot	93.244
HAR__13	tBodyAcc.Jerk-mean()-Y	Violin plot	95.780
HAR__14	tBodyAcc.Jerk-mean()-Z	Histogram	94.890
HAR__15	tBodyAcc.Jerk-std()-X	Dot plot	95.703
HAR__16	tBodyAcc.Jerk-std()-Y	Heatmap	96.787
HAR__17	tBodyAcc.Jerk-std()-Z	Dot plot	91.893
HAR__18	tBodyGyro-mean()-X	Dot plot	93.936
HAR__19	tBodyGyro-mean()-Y	Violin plot	85.860
HAR__20	tBodyGyro-mean()-Z	Heatmap	93.689
HAR__21	tBodyGyro-std()-X	Violin plot	90.400
HAR__22	tBodyGyro-std()-Y	Dot plot	95.275
HAR__23	tBodyGyro-std()-Z	Histogram	93.545
HAR__24	tBodyGyro.Jerk-mean()-X	Heatmap	97.003
HAR__25	tBodyGyro.Jerk-mean()-Y	Histogram	95.208
HAR__26	tBodyGyro.Jerk-mean()-Z	Histogram	95.146
HAR__27	tBodyGyro.Jerk-std()-X	Heatmap	95.374
HAR__28	tBodyGyro.Jerk-std()-Y	Dot plot	94.850
HAR__29	tBodyGyro.Jerk-std()-Z	Histogram	94.960
HAR__30	tBodyAccMag-mean()	Dot plot	93.460
HAR__31	tBodyAccMag-std()	Histogram	94.799
HAR__32	tGravityAccMag-mean()	Dot plot	93.460
HAR__33	tGravityAccMag-std()	Histogram	94.799
HAR__34	tBodyAcc.JerkMag-mean()	Heatmap	94.984
HAR__35	tBodyAcc.JerkMag-std()	Heatmap	94.458
HAR__36	tBodyGyroMag-mean()	Heatmap	96.859
HAR__37	tBodyGyroMag-std()	Dot plot	89.664
HAR__38	tBodyGyro.JerkMag-mean()	Histogram	91.473
HAR__39	tBodyGyro.JerkMag-std()	Heatmap	90.191
HeartFailure__00	age	Heatmap	96.273
HeartFailure__01	creatinine_phosphokinase	Histogram	97.494
HeartFailure__02	ejection_fraction	Heatmap	96.308
HeartFailure__03	platelets	Dot plot	97.795
HeartFailure__04	serum_creatinine	Dot plot	96.079
HeartFailure__05	serum_sodium	Violin plot	91.515
ILPD__00	age	Histogram	96.813
ILPD__01	total_bilirubin	Heatmap	96.342
ILPD__02	direct_bilirubin	Heatmap	96.157
ILPD__03	alkaline_phosphatase	Heatmap	97.540
ILPD__04	ALT	Heatmap	98.177
ILPD__05	AST	Heatmap	96.363
ILPD__06	total_protein	Dot plot	97.163
ILPD__07	albumin	Histogram	95.607
ILPD__08	albumin_globulin_ratio	Heatmap	95.833
Retinopathy__00	feature_2	Heatmap	98.481
Retinopathy__01	feature_3	Dot plot	97.699
Retinopathy__02	feature_4	Heatmap	97.717
Retinopathy__03	feature_5	Histogram	97.775
Retinopathy__04	feature_6	Dot plot	97.731
Retinopathy__05	feature_7	Dot plot	96.111
Retinopathy__06	feature_8	Heatmap	98.142
Retinopathy__07	feature_9	Heatmap	99.283
Retinopathy__08	feature_10	Heatmap	99.338
Retinopathy__09	feature_11	Heatmap	98.663
Retinopathy__10	feature_12	Heatmap	93.375
Retinopathy__11	feature_13	Heatmap	92.412
Retinopathy__12	feature_14	Heatmap	92.812

Key	Feature	Winning family set	Score
Retinopathy__13	feature_15	Heatmap	95.556
Retinopathy__14	feature_16	Histogram	98.066
Retinopathy__15	feature_17	Heatmap	98.090
WDBC__00	mean radius	Heatmap	99.023
WDBC__01	mean texture	Histogram	97.103
WDBC__02	mean perimeter	Heatmap	97.848
WDBC__03	mean area	Dot plot	97.698
WDBC__04	mean smoothness	Dot plot	98.113
WDBC__05	mean compactness	Dot plot	97.838
WDBC__06	mean concavity	Heatmap	97.843
WDBC__07	mean concave points	Heatmap	96.921
WDBC__08	mean symmetry	Dot plot	98.493
WDBC__09	mean fractal dimension	Dot plot	98.391
WDBC__10	radius error	Heatmap	97.981
WDBC__11	texture error	Dot plot	97.868
WDBC__12	perimeter error	Heatmap	98.585
WDBC__13	area error	Dot plot	94.793
WDBC__14	smoothness error	Dot plot	97.963
WDBC__15	compactness error	Heatmap	98.883
WDBC__16	concavity error	Heatmap	98.241
WDBC__17	concave points error	Heatmap	97.423
WDBC__18	symmetry error	Heatmap	97.941
WDBC__19	fractal dimension error	Dot plot	97.445
WDBC__20	worst radius	Heatmap	98.920
WDBC__21	worst texture	Heatmap	97.953
WDBC__22	worst perimeter	Heatmap	98.816
WDBC__23	worst area	Dot plot	97.499
WDBC__24	worst smoothness	Dot plot	97.516
WDBC__25	worst compactness	Dot plot	97.668
WDBC__26	worst concavity	Heatmap	98.677
WDBC__27	worst concave points	Heatmap	97.531
WDBC__28	worst symmetry	Heatmap	97.011
WDBC__29	worst fractal dimension	Violin plot	97.566
Wrist__00	mean	Bar chart	95.348
Wrist__01	median	Bar chart	95.645
Wrist__02	q10	Violin plot	94.784
Wrist__03	q90	Bar chart	95.484
Wrist__04	SD	Violin plot	90.352

A best score has limits

Resolution sensitivity and the lowest-scoring primary optimum



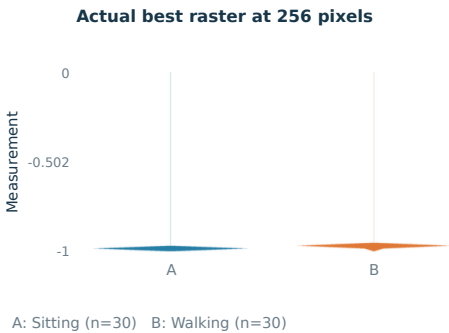
b Lowest reference optimum

RANK / CHART AND SETTING		SCORE / 100
1	Violin plot 64 grid points	81.015
2	Dot plot raw observations	80.257
3	Heatmap 64 bins	67.815
4	Interval plot five-number	65.805
5	Histogram 64 bins	64.582
6	Bar chart means	54.992
-	Box plot five-number	Decode failed

Higher is better • Green: computational optimum

Activity / Gravity variability, x axis

Best score: 81.015 / 100



A lower score can reflect summary loss, rendering or decoder limitations. It does not diagnose poor source data.

Figure S1: **Rendering robustness and the lowest-scoring reference optimum.** The former main Figure 3 is retained here to distinguish resolution sensitivity from limited gain over a strong baseline. Winner sets change for 77 of 127 primary comparisons between 128 and 512 pixels. The lowest reference optimum is 81.015 for HAR gravity-acceleration variability. All original scores and failures remain unchanged. This analysis does not establish a source-data ceiling on human performance.

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